

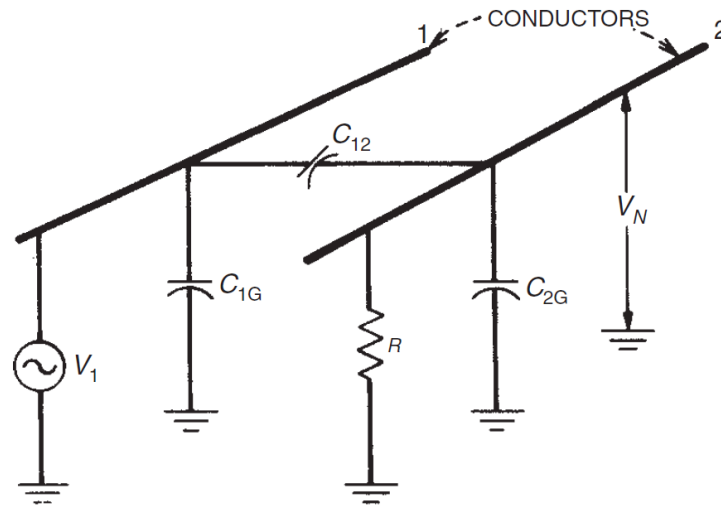
Exercise 1 (EMI/EMC) (14%)


Figure 1. Diagram of two conductors with capacitive coupling

A diagram of two conductors with capacitive coupling is shown in Figure 1. With an assumption that conductors are short compared with a wavelength:

Question A (6%). Draw the equivalent circuit diagram and calculate the noise voltage (V_N) picked up by the second conductor.

Question B (6%). Consider the parasitic capacitances and the noise source given as below:

V_1 (noise source) = 10 Vac @ 100 kHz
$C_{12} = 50$ pF
$C_{1G} = C_{2G} = 150$ pF

Calculate V_N if the termination resistance R is:

Question B.1. $R = 100$ k Ω

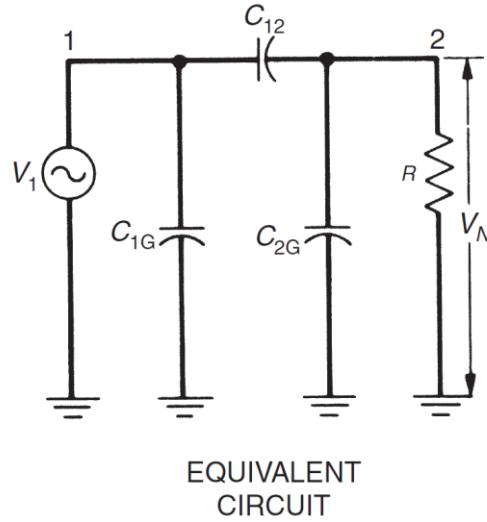
Question B.2. $R = 50$ Ω

Question C (2%). Based on the calculated noise voltage (V_N) in Part A and the obtained results in Part B, how does the termination resistance R minimize the noise voltage (V_N) picked by the second conductor?

Solution

Question A.

The equivalent circuit of Figure 1 is: (1%)



From the obtained equivalent circuit V_N is calculated from V_1 based on voltage division principle:

(3%)

$$V_N = \frac{Z_2}{Z_1 + Z_2} V_1$$

$$Z_1 = \frac{1}{j\omega C_{12}}$$

$$Z_2 = \frac{1}{j\omega C_{2G}} \parallel R = \frac{R}{j\omega R C_{2G} + 1}$$

$$\Rightarrow V_N = \frac{\frac{R}{j\omega R C_{2G} + 1}}{\frac{R}{j\omega R C_{2G} + 1} + \frac{1}{j\omega C_{12}}} V_1 = \frac{j\omega R C_{12}}{j\omega R (C_{12} + C_{2G}) + 1} V_1$$

EQUIVALENT
CIRCUIT

$$\Rightarrow V_N = \frac{\frac{R C_{12}}{(C_{12} + C_{2G})}}{R + \frac{1}{j\omega (C_{12} + C_{2G})}} V_1$$

Condition 1: $R \ll \frac{1}{j\omega (C_{12} + C_{2G})} \longrightarrow V_N = j\omega R C_{12} V_1$ (1%)

Condition 2: $R \gg \frac{1}{j\omega (C_{12} + C_{2G})} \longrightarrow V_N = \left(\frac{C_{12}}{C_{12} + C_{2G}} \right) V_1$ (1%)

Question B.

In order to calculate the noise voltage picked by second conductor, considering the giving parameter values the following term should be calculated:

$$\frac{1}{j\omega(C_{12} + C_{2G})} \xrightarrow[C_{12}=50\text{ pF}, C_{2G}=150\text{ pF}]{\omega=2\pi \times 100 \times 10^3} \left| \frac{1}{j\omega(C_{12} + C_{2G})} \right| \approx 7.96\text{ k}\Omega \quad (\text{I}) \quad (2\%)$$

B.1. Considering the calculated the ratio above (I) and termination resistance $R = 100\text{ k}\Omega$, then V_N can be estimated following condition 2

$$V_N = \left(\frac{C_{12}}{C_{12} + C_{2G}} \right) V_1 \longrightarrow V_N = \frac{50}{200} \times 10 = 2.5\text{ V} \quad (2\%)$$

B.2. Considering the calculated the ratio above (I) and termination resistance $R = 50\text{ }\Omega$, then V_N can be estimated following condition 1

$$V_N = j\omega RC_{12} V_1 \longrightarrow |V_N| = 2 \times \pi \times 100 \times 10^3 \times 50 \times 50 \times 10^{-12} \times 10 = 15.7\text{ mV} \quad (2\%)$$

Question C. (2%)

Considering the above calculations, the termination resistance can minimize the noise voltage (V_N) picked by the second conductor if condition 1 is hold.

This means that the termination resistance should be

$$R \ll \frac{1}{j\omega(C_{12} + C_{2G})} \longrightarrow V_N = j\omega RC_{12} V_1$$

From the above relation it is obvious that reducing R can directly reduce V_N as well. Ideally if R is zero, V_N will be zero as well (grounding the second conductor $R = 0$).